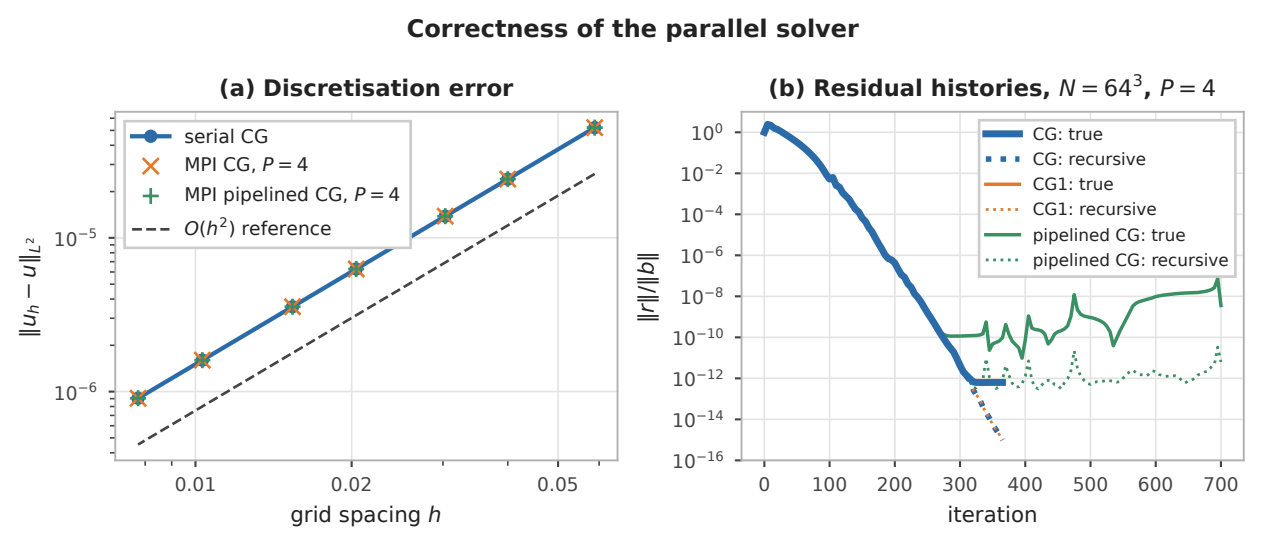
Correctness of the parallel solver
(a) Discretisation error
serial CG
MPI CG, $P = 4$
MPI pipelined CG, $P = 4$
$O(h^2)$ reference
$\|u_h - u\|_{L^2}$
10^{-5}
10^{-6}
0.01
0.02
0.05
grid spacing h
(b) Residual histories, $N = 64^3$, $P = 4$
CG: true
CG: recursive
CG1: true
CG1: recursive
pipelined CG: true
pipelined CG: recursive
$\|r\|/\|b\|$
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
0
100
200
300
400
500
600
700
iteration